Do Now · retrieval

Database tables, entities and keys

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Define INNER JOIN.

Q2

What is the purpose of a primary key?

Q3

Which SQL clause filters rows?

Q4

Insert 12 into BST with root 10 and right child 15. State its final position.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Understand — An entity corresponds to a distinguishable object type; a relational table stores rows of its attributes.

Core idea 2

Apply — A primary key uniquely identifies each row; foreign keys express references from related tables.

Core idea 3

Evaluate — A secondary/index key can accelerate searching without necessarily being unique. One-to-many relationships use a foreign key on the many side.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

entity — Type of identifiable object.

attribute — Descriptive field of an entity.

primary key — Unique row identifier.

foreign key — Reference to key in another table.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

Identify primary and foreign keys in Customer(CustomerID, Name) and Order(OrderID, CustomerID).

Follow the worked method

1. Predict first — Orders have their own identity and store a reference to the customer who placed them.

2. Model answer — Explain the reasoning and justify the output.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict first

The Homework many-side contains PupilID referencing the one-side table.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Define entity, attribute, primary key, foreign key. [4]

Question 2

For Student, Book, Loan given above, state PKs and both FKs. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Explain why storing every loan and every pupil detail in a single repeated row causes anomalies. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

A foreign key must always be unique across every row in its table.

Use key vocabulary from memory

1. Explain entity in your own words.

2. Explain attribute in your own words.

3. Explain primary key in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.